

Muhammad Danial Aslam

Islamabad, Pakistan | danialaslam04@gmail.com | +92 305 5250224 | LinkedIn | GitHub | Portfolio

Summary

Software engineer with three years of experience designing and delivering production backend systems and agentic AI applications on Google Cloud. Skilled in translating complex business requirements into production-grade software, working across Java/Spring Boot, Trillo Workbench, and Python/FastAPI to build REST APIs, async workflows, and third-party integrations. Currently specialising in agentic AI engineering — building agentic systems with Google ADK, Model Context Protocol (MCP), and Vertex AI, where models propose and deterministic code executes behind governed tool boundaries. Experienced in working directly with clients to turn raw operational requirements into domain models and shipped software.

Experience

Software Engineer, Z-Axis SMC Pvt Ltd – Islamabad, Pakistan

October 2023 – present

- Engineered backend and agentic-systems across multiple production platforms, taking projects from direct client engagement and requirements translation through domain modelling, API design, and production delivery.
- Designed and built backend services for a business-acquisition marketplace serving buyers, brokers, and administrators across a Java/Trillo API/Spring Boot domain backend and a Python/FastAPI agentic service running five LLM agents on Google ADK, with Stripe subscription billing and Plaid-backed verification.
- Built the backend for a concierge care platform connecting aging-adult clients with field staff, including an onboarding agent, a task-planning agent, and a pod matching engine, integrated with Gusto Embedded Payroll and Checkr background checks.
- Developed backend services for a multi-tenant drayage TMS automating the order-to-delivery lifecycle, including an LLM pipeline that ingests delivery orders from email, extracts and validates structured data, runs credit checks, and creates loads automatically.
- Applied one agent-execution pattern across platforms: MCP tool servers with per-call identity propagation, role-filtered toolsets bounding which tools each agent can call, and a risk-based gate holding high-risk writes for explicit approval.
- Built on Google Cloud using Vertex AI and Gemini for model inference, Cloud SQL and Cloud Storage for persistence, Secret Manager for credentials, and BigQuery vector search for semantic discovery, with services running on GKE and Redis-backed caching and rate limiting.

Data Science Intern, Soltridge – Islamabad, Pakistan

May 2023 – July 2023

- Developed a house-price prediction model for the Pakistani market using Python and Scikit-Learn, achieving 91% accuracy on property and macro-economic features.
- Web-scraped and assembled the housing dataset from zameen.com, performing data preprocessing, exploratory data analysis, and feature engineering.

Skills

Agentic AI: Google ADK, LangGraph, Model Context Protocol (MCP), Multi-Agent Systems, Agent Orchestration, Tool Calling

LLM & Gen AI: Vertex AI, Gemini, Claude, LiteLLM, Prompt Engineering, Embeddings, Retrieval-Augmented Generation (RAG)

Languages: Java, Python, SQL

Frameworks: Trillo Workbench, FastAPI, Spring Boot

Cloud Computing: Google Cloud Platform (GCP)

Tools: Git, GitHub, Postman, Jira

Databases & Infrastructure: PostgreSQL, pgvector, MySQL, Redis, Docker, Kubernetes (GKE)

Soft Skills: Client Engagement, Requirements Discovery, Cross-functional Collaboration, Stakeholder Communication, Technical Communication, Ownership, Adaptability

Education

Bachelor of Science in Computer Science

International Islamic University Islamabad • CGPA: 3.31 • 2023

Intermediate in Computer Science

Punjab College, Blue Area, Islamabad • Grade: B • 2018

Certifications

Introduction to Large Language Models

Google Cloud

Introduction to Generative AI

Google Cloud

Introduction to Responsible AI

Google Cloud

Computer Vision Fundamentals with Google Cloud

Google Cloud

Recommendation Systems on Google Cloud

Google Cloud

Web Development Course

IQRA University Islamabad